The Kirchhoff Šoltés problem: non-existence theorems and exhaustive searches

chemistry-open-problems · 24 October 2026 · preliminary (draft by the solver)

Solver: Claude Opus 5.5 (effort xhigh) · Verification: pending

Abstract. Gottwald, Majstorović Ergotić and Došlić asked for all graphs G with $\mathrm{Kf}(G-v) = \mathrm{Kf}(G)$ for every vertex v (Kirchhoff index = sum of effective resistances); only the cycle C_5 is known. We prove non-existence for several classes and extend the published searches. Along a path of degree-two vertices, $\mathrm{Kf}(G-v_i)$ is a strictly convex quadratic in the position i; hence a solution other than C_5 with cyclomatic number c has at most $8(c-1)$ vertices, and a kernel enumeration shows that there is none with $c \le 5$. Every solution satisfies $a(G)^2 \le 2n/\sum_v d_v^{-1}$ (a the algebraic connectivity). Complete multipartite graphs, strongly regular graphs other than C_5, triangle-free distance-regular graphs other than C_5, distance-regular graphs of valency at most four other than C_5, all hypercubes and all Hamming graphs are not solutions; for triangle-free distance-regular graphs we give the closed form $\mathrm{Kf}(G)-\mathrm{Kf}(G-v) = [(k-1)(n-1)^2-2k(n-k-1)\,\mathrm{Kf}(G)/n]/[k((k-2)n+2)]$. An exact modular criterion shows that C_5 is the only solution on at most 11 vertices (910 914 360 two-connected graphs) and that there is no vertex-transitive solution on $n \le 39$ or $n \in \{41, 43, 45, 47\}$ vertices (Holt–Royle census). The problem remains open.

The question

Source [1], p. 5 of arXiv:2609.13751:

> *"Problem 3. Find all graphs G so that the equality $Kf(G) = Kf(G-v)$ holds for all $v \in V(G)$."*

and p. 9: "Our numerical studies failed to identify any additional Kirchhoff Šoltés graphs. We investigated connected graphs with up to 9 vertices. [...] we examined all vertex-transitive graphs with up to 23 vertices [...]. None of these graphs were Kirchhoff Šoltés." The problem is the resistance-distance analogue of Šoltés's problem for the Wiener index [2] [3] [4] [5] [6] [7]; a relaxed version is studied in [8].

Conventions. $\mathrm{Kf}(G) = \sum_{u<w} r_G(u,w) = n\sum_{i=1}^{n-1} 1/\mu_i$ [9] [10], where $\mu_1 \ge \cdots \ge \mu_{n-1} > 0$ are the non-zero Laplacian eigenvalues; $a(G) = \mu_{n-1}$; $\Gamma = L(G)^+$; $\mathrm{Rt}_G(v) = \sum_u r_G(u,v)$; $\tau(G)$ is the number of spanning trees; $\Delta_v(G) = \mathrm{Kf}(G) - \mathrm{Kf}(G-v)$. As in the source, v is *good* if $G-v$ is connected and $\Delta_v = 0$, and a *Kirchhoff Šoltés (KS) graph* is a connected graph on $n \ge 3$ vertices all of whose vertices are good. Every KS graph is 2-connected.

1 Tools

Lemma 1 (grounded formula)**.** Let H be connected on N vertices, $g \in V(H)$, A the Laplacian of H with the row and column of g deleted and $M = A^{-1}$. Then $\det A = \tau(H)$ and $\mathrm{Kf}(H) = N\,\mathrm{tr}\,M - \mathbf{1}^{\mathsf{T}} M \mathbf{1}$; in particular $\tau(H)\,\mathrm{Kf}(H) \in \mathbb{Z}$.

Proof. M is the Green function grounded at g, so $r(u,g) = M_{uu}$ and $r(u,w) = M_{uu} + M_{ww} - 2M_{uw}$ for $u, w \neq g$. Summing over pairs gives $(N-2)\operatorname{tr} M - (\mathbf{1}^{\mathsf{T}} M \mathbf{1} - \operatorname{tr} M) + \operatorname{tr} M$. Integrality: $M = \operatorname{adj}(A)/\det A$. □

We use Rayleigh monotonicity, $r_H(x,y) \leq d_H(x,y)$, Kron reduction (eliminating a vertex v of degree d replaces its star by a complete graph on $N(v)$ with conductances $1/d$) and the Woodbury identity.

2 Threads and bounded cyclomatic number

Lemma 2 (thread formula). Let G be connected and $P = xv_1 \cdots v_{\ell-1} y$ ($\ell \geq 2$) a path whose internal vertices have degree 2, $x \neq y$, and $R = G - \{v_1, \ldots, v_{\ell-1}\}$ connected with n_R vertices. For $a = i-1$, $b = \ell - 1 - i$,

$$\operatorname{Kf}(G - v_i) = \operatorname{Kf}(R) + n_R\left[\tbinom{a+1}{2} + \tbinom{b+1}{2}\right] + a\operatorname{Rt}_R(x) + b\operatorname{Rt}_R(y) + \frac{a^3-a}{6} + \frac{b^3-b}{6} + \frac{ab(a+b+2)}{2} + ab\, r_R(x,y).$$

For fixed $s = a + b$ this is a polynomial of degree ≤ 2 in a with leading coefficient $n_R - 1 - r_R(x,y)$.

Proof. $G - v_i$ is R with pendant paths of a vertices at x and b vertices at y; pendant paths hang on bridges, so all resistances involving them are sums. Summing the six types of pairs gives the formula. With $b = s - a$: $n_R[a(a+1) + b(b+1)]/2$ contributes $n_R a^2$, $(a^3 + b^3)/6$ contributes $(s/2)a^2$, and $ab = sa - a^2$ contributes $-((s+2)/2 + r_R(x,y))a^2$. □

Lemma 3. For connected R and $x \neq y$, $r_R(x,y) \leq n_R - 1$, with equality iff R is a path with ends x and y.

Proof. $r_R(x,y) \leq d_R(x,y) \leq n_R - 1$. If equality holds a shortest x–y path P' is Hamiltonian, and any further edge $p_i p_j$ ($j \geq i+2$) puts a unit resistor in parallel with a segment of resistance $j - i$, giving $r < n_R - 1$. □

Corollary 4. Let G be 2-connected but not a cycle. At most two internal vertices of any thread (maximal path of degree-two vertices) are good; if a thread has exactly two internal vertices and both are good then $\operatorname{Rt}_R(x) = \operatorname{Rt}_R(y)$. Hence a KS graph is C_5 or has all threads with at most two internal vertices.

Proof. The ends $x \neq y$ of a thread have degree ≥ 3, and R is connected (otherwise x would be a cut vertex); R is not an x–y path. By Lemmas 2 and 3, $a \mapsto \operatorname{Kf}(G - v_{a+1})$ is a strictly convex quadratic, which takes the value $\operatorname{Kf}(G)$ at most twice. For $\ell = 3$ the two values differ by $\operatorname{Rt}_R(x) - \operatorname{Rt}_R(y)$. A KS cycle C_n needs $(n^3 - n)/12 = (n-2)(n-1)n/6$, i.e. $n = 5$ [1]. □

Theorem 5. A KS graph $G \neq C_5$ with cyclomatic number $c = m - n + 1$ satisfies $c \geq 2$ and $n \leq 8(c-1)$, and is a subdivision, with 0, 1 or 2 new vertices per edge, of a loopless multigraph without cut vertex, of minimum degree ≥ 3, with $n_K \leq 2(c-1)$ vertices and $m_K \leq 3(c-1)$ edges. There is no such graph with $c \leq 5$; hence every KS graph other than C_5 has $m \geq n + 5$.

Proof. The kernel (suppress degree-two vertices) is loopless, has no cut vertex and the same cyclomatic number; $2m_K \geq 3n_K$ and Corollary 4 give the bounds. The kernels are enumerated with nauty [11] (`geng -C` for the underlying simple graph, `multig -e` for multiplicities) and every admissible subdivision is tested with the exact criterion of Section 7: $c = 2$: 7 graphs (the theta graphs with path lengths in $\{1, 2, 3\}$); $c = 3$: 4 kernels, 1 038 graphs; $c = 4$: 17 kernels, 76 603 graphs; $c = 5$: 118 kernels, 7 665 126 graphs ($5 \leq n \leq 32$). None is KS. □

3 An expander bound and a dense bound

For $u \neq w$ let $\varphi = \Gamma(e_u - e_w)$ and $E_v(u, w) = \sum_{x \sim v} (\varphi_x - \varphi_v)^2$.

Lemma 6. If G and $H = G - v$ are connected and $a_v = a(H)$, then for $u, w \in V(H)$

$$E_v(u, w) \leq r_H(u, w) - r_G(u, w) \leq (1 + 1/a_v)\, E_v(u, w).$$

Proof. Kron reduction gives $L_{H'} = L_H + S$ with $S = P_N - \frac{1}{d}\mathbf{1}_N\mathbf{1}_N^{\mathsf{T}}$, the orthogonal projector onto $Y_N = \{y : \operatorname{supp} y \subseteq N(v),\ \sum y = 0\}$; write $S = QQ^{\mathsf{T}}$. On $\mathbf{1}^{\perp}$, $X = L_{H'}^{-1}$ and Woodbury give $r_H - r_G = c^{\mathsf{T}}(I - K)^{-1}c$, $c = Q^{\mathsf{T}}Xb$, $K = Q^{\mathsf{T}}XQ$. From $L_H \succeq a_v I$ we get $L_H + S \succeq a_v I + S$, which acts as $(1 + a_v)$ on Y_N, so $0 \preceq K \preceq (1 + a_v)^{-1} I$ and $I \preceq (I - K)^{-1} \preceq (1 + 1/a_v) I$. Finally $\|c\|^2 = \sum_{x \sim v} (\varphi_x - \varphi_v)^2$ because φ is harmonic at v. □

Theorem 7 (expander bound). Let G be 2-connected, $a_* = \min_v a(G - v)$ and $\varepsilon_v = \sum_{w \neq v} E_v(v, w) \geq (n - 1)/d_v$. Then

$$\sum_v \Delta_v(G) \;\geq\; (1 + 1/a_*) \sum_v \varepsilon_v - 2\,\mathrm{Kf}(G)/a_* \;\geq\; \tfrac{1}{a_*}\Big[a(G)(n-1)\sum_v d_v^{-1} - 2\,\mathrm{Kf}(G)\Big].$$

Hence G is not KS if $a(G)(n - 1)\sum_v d_v^{-1} > 2\,\mathrm{Kf}(G)$, in particular if $a(G)^2 > 2n/\sum_v d_v^{-1}$; if G is vertex-transitive, no vertex is good.

Proof. $\Delta_v = \mathrm{Rt}_G(v) - \sum_{u<w;\, u,w \neq v} (r_{G-v} - r_G)$, bounded with Lemma 6. One checks $\sum_{u<w} E_v(u, w) = n \operatorname{tr}(\Gamma^2 L_v^{\star})$ with $L_v^{\star}$ the Laplacian of the star at v, and $\sum_v L_v^{\star} = 2L$, so these sums add up to $2\,\mathrm{Kf}(G)$. The edge currents at v sum to 1, which gives $\varepsilon_v \geq (n - 1)/d_v$. Further $1 + a_* \geq a(G)$ (Fiedler [12]: extend a Fiedler vector of $G - v$ by 0) and $\mathrm{Kf}(G) \leq n(n - 1)/a(G)$. □

Corollary 8. Every graph with $a(G)^2 > 2(n-1)$ is not KS; in particular graphs with minimum degree $\delta > (n - 2 + \sqrt{2(n-1)})/2$ (since $a \geq 2\delta - n + 2$), joins $H_1 + H_2$ with $\min(n_1, n_2) > \sqrt{2(n-1)}$, Johnson graphs, Paley graphs $P(q)$ with $q \geq 9$, and d-regular graphs with second eigenvalue $\leq 2\sqrt{d-1}$ and $d \geq 12$.

Theorem 9 (dense bound). If $a(G) > 1$ then for every v, $\Delta_v(G) \geq 1 - \sum_{i=1}^{n-1} \frac{n - \mu_i}{\mu_i(\mu_i - 1)}$. The bound is attained by the graphs $K_s + K_{s+1,1}$ of [1] (half of whose vertices are good).

Proof. With $\Phi(F,N)=\sum_{\theta\in\operatorname{spec}L(F)}\theta/(N-\theta)$, $\mathrm{Kf}(G)=n-1+\Phi(\bar G,n)$ and $\mathrm{Kf}(G-v)=n-2+\Phi(\bar G-v,n-1)$. Deleting the edges of $\bar G$ at v lowers every eigenvalue (Weyl), and $\theta/(n-1-\theta)$ is increasing on $[0,n-1)$; so $\Delta_v\ge 1-\sum_\theta\theta/((n-\theta)(n-1-\theta))$ with $\theta=n-\mu$. □

4 A local deletion formula

Proposition 10. Let v be a non-cut vertex, $N=N(v)$, $d=|N|$, $\Gamma_2=\Gamma^2$, and for $a,b\in N$: $M_{ab}=\Gamma_{ab}-\Gamma_{av}-\Gamma_{vb}+\Gamma_{vv}$, $(M^2)_{ab}=(\Gamma_2)_{ab}-(\Gamma_2)_{av}-(\Gamma_2)_{vb}+(\Gamma_2)_{vv}+n(\Gamma_{av}-\Gamma_{vv})(\Gamma_{bv}-\Gamma_{vv})$, $s_a=n(\Gamma_{vv}-\Gamma_{av})$, $t_a=-n[(\Gamma_2)_{av}-(\Gamma_2)_{vv}]-n^2\Gamma_{vv}(\Gamma_{av}-\Gamma_{vv})$,

$$Z=\begin{pmatrix} M_{NN}-I & s\\ s^{\mathsf T} & n-1+n^2\Gamma_{vv}\end{pmatrix},\qquad W=\begin{pmatrix} M_{NN}^2 & t\\ t^{\mathsf T} & n^2(\Gamma_2)_{vv}+n^3\Gamma_{vv}^2\end{pmatrix}.$$

Then $\mathrm{Kf}(G-v)=(n-1)[\mathrm{Rt}_G(v)-1-\operatorname{tr}(Z^{-1}W)]$ with $\mathrm{Rt}_G(v)=\operatorname{tr}\Gamma+n\Gamma_{vv}$.

Proof. With $M=L_v^{-1}$ (grounded at v), $L_{G-v}=M^{-1}-P_N$ and $\mathrm{Kf}(G-v)=(n-1)(\operatorname{tr}Y^{-1}-1)$ for $Y=L_{G-v}+J/(n-1)=M^{-1}+UCU^{\mathsf T}$, $U=[E_N,\mathbf 1]$, $C=\operatorname{diag}(-I_d,1/(n-1))$; apply Woodbury and express M, M^2 through Γ. □

For a distance-regular graph with intersection array $\{b_0,\dots;c_1,\dots\}$, Γ and Γ^2 are radial and follow from the layer equations $k_ib_i(\psi_i-\psi_{i+1})=\sum_{l\le i}k_lf_l$ [13]; two neighbours of v are at distance 1 or 2, so $\operatorname{tr}(Z^{-1}W)$ splits over the eigenvalues of the local graph (`scripts/drg_formula.py`, validated against direct exact computation on 17 graphs).

5 Complete multipartite and strongly regular graphs

Theorem 11. For $G=K_{n_1,\dots,n_k}$ (not a star) and v in a part of size n_j, $\Delta_v=\phi(n_j)-S$ with $\phi(x)=\frac{n+x-2}{n-x}+\frac{x(x-1)}{(n-x)(n-1-x)}$ strictly increasing and $S=\sum_i\frac{n_i(n_i-1)}{(n-n_i)(n-1-n_i)}$. The good vertices form the union of the parts of one size, and G is never KS: for $K_{k\times s}$, $\Delta_v=\frac{k(k-1)s^2-2ks+2}{(k-1)s((k-1)s-1)}>0$.

Theorem 12. C_5 is the only strongly regular KS graph. For a triangle-free $\mathrm{SRG}(n,k,0,\mu)$,

$$\Delta_v=\frac{k^4+3k^3\mu-5k^3+3k^2\mu^2-9k^2\mu+5k^2+k\mu^3-3k\mu^2+5k\mu-k+\mu^3-2\mu^2+\mu}{\mu(k+\mu-2)(k^2+k\mu-k+\mu)}>0$$

unless $(k,\mu)=(2,1)$; for Moore graphs of diameter two $\Delta_v=k(k-2)(k+1)/(k^2+1)$.

Proof. Imprimitive SRGs are complete multipartite (Theorem 11). Conference graphs with $n\ge 9$ have $a^2=((n-\sqrt n)/2)^2>2k$, and primitive SRGs with integral eigenvalues $r\ge 1$, $s\le -2$ have $\mu=k+rs\le k-2r$, hence $a=k-r\ge(k+1)/2$ and $a^2>2k$ for $k\ge 6$; both are excluded by Theorem 7. The feasible primitive parameter sets with $k\le 5$ are $(5,2,0,1)$, $(9,4,1,2)$, $(10,3,0,1)$, $(16,5,0,2)$; the closed form (and, for $(9,4,1,2)$, Jensen and chord bounds on the local-graph eigenvalue sum, $\Delta\in[64/45,8/5]$) settles them. □

6 Distance-regular graphs

Theorem 13. (i) If G is distance-regular with $a_1 = 0$, valency $k \geq 2$ and diameter ≥ 2, then at every vertex

$$\mathrm{Kf}(G) - \mathrm{Kf}(G - v) = \frac{(k-1)(n-1)^2 - 2k(n-k-1)\,\mathrm{Kf}(G)/n}{k((k-2)n+2)}\,.$$

(ii) C_5 is the only KS triangle-free distance-regular graph; for valency ≥ 5 no vertex is good. (iii) C_5 is the only KS distance-regular graph of valency at most four. (iv) No vertex of a hypercube Q_d or of a Hamming graph $H(d,q)$ $(d,q \geq 2)$ is good. (v) A distance-regular graph of valency ≥ 3 with $a(G) \geq 2(1 + 94/101)$ is not KS.

Proof. (i) is Proposition 10 with $b_1 = k - 1$; the Γ^2 terms cancel. (ii) For $k \geq 5$, Markowsky and Koolen Theorem 1 [14] give $r_D \leq K r_1$, $K = 1 + 94/101$, hence $\mathrm{Kf} \leq K(n-1)^2/k$, below the threshold $n(k-1)(n-1)^2/(2k(n-k-1))$ because $2K < 4 \leq k-1$. For $k = 3, 4$ use the classification lists [15] [16] quoted in Cases 2–3 [14] and exact evaluation; $k = 2$ gives $\Delta(C_n) = -n(n-1)(n-5)/12$. (iii) The same lists, including the four quartic graphs with $a_1 = 1$. (iv) $\Delta(Q_d) > 0 \iff (d-1)(n-1)^2 > d(n-d-1)S_1$, $S_1 = \sum_i \binom{d}{i}/i < 4n/(d+1)$; for $H(d,q)$ the analogous closed form is linear in Kf and $S_q = \sum_i \binom{d}{i}(q-1)^i/i < 2qn/((d+1)(q-1))$ suffices except for finitely many cases computed exactly. (v) Combine Theorem 7 with $\mathrm{Kf} \leq K(n-1)^2/k$. □

Closed forms valid for all parameters are also obtained for incidence graphs of symmetric designs, of generalized quadrangles and hexagons of order (q,q) and for crown graphs (`check_drg_symbolic.py`).

7 Exhaustive searches

Exact criterion. For $H \in \{G, G - v\}$, $k_p(H) = N \operatorname{tr} A^{-1} - \mathbf{1}^{\mathsf{T}} A^{-1} \mathbf{1} \bmod p$ (Lemma 1, $\mathrm{LDL}^{\mathsf{T}}$ over $\mathbb{F}_p$) equals $\tau(H)\,\mathrm{Kf}(H) \cdot \tau(H)^{-1}$; if $k_p(G) \neq k_p(G - v)$ then $p \nmid \tau(G - v)\tau(G)\,\mathrm{Kf}(G) - \tau(G)\tau(G - v)\,\mathrm{Kf}(G - v)$, so $\mathrm{Kf}(G) \neq \mathrm{Kf}(G - v)$. Graphs passing every vertex for two primes are re-checked in exact rationals. Implementations: `kscheck.c` ($p =$ 1073741789, 1073741783) and `kscheck_fast.c` ($p =$ 268435399, 268435367), validated against exact rational values (2 126 Kirchhoff indices).

Theorem 14. C_5 is the only KS graph with at most 11 vertices (all 910 914 360 two-connected graphs on $3 \leq n \leq 11$ vertices generated by `geng -C`; counts equal to OEIS A002218).

Theorem 15. There is no vertex-transitive KS graph on n vertices for $6 \leq n \leq 39$ and $n \in \{41, 43, 45, 47\}$ (every graph of the corresponding files of the Holt–Royle census [17] [18], all valencies; counts per n equal to OEIS A006799). The files $n = 40, 42, 44, 46$ were not processed.

8 Limitations

No KS graph other than C_5 was found; the general problem remains open. The searches are exact but finite; floating-point scans of larger families (lattice quotients of $\mathbb{Z}^2$ up to 1000 vertices, generalized Petersen graphs up to $GP(1000, k)$) are explorations only and

are not used in any statement. The valency-3 and 4 classification of distance-regular graphs is used as quoted in [14].

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